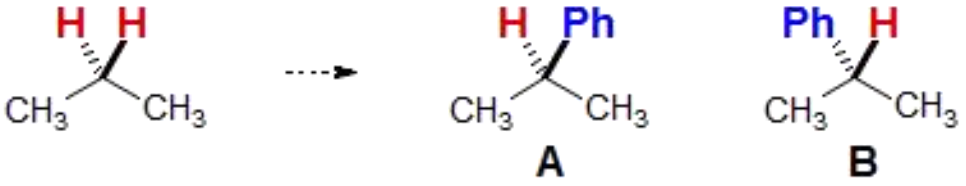
第三章　分子的对称性和群论基础

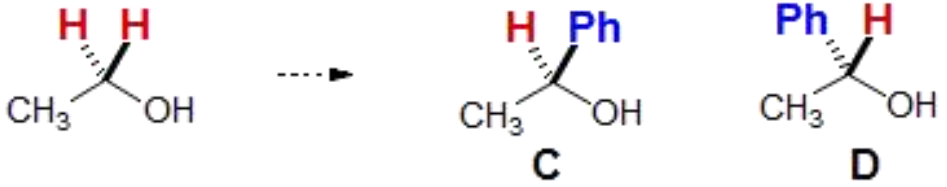
- 对称操作和对称元素怎样数学表达？
- 什么是群？
- 什么是特征标？怎样用特征标表达对称性？
- 怎么利用特征标表理解红外光谱？

群论在化学中应用的核心很简单：
判断积分是否为零
高级版的“奇函数积分为零，偶函数不为零”

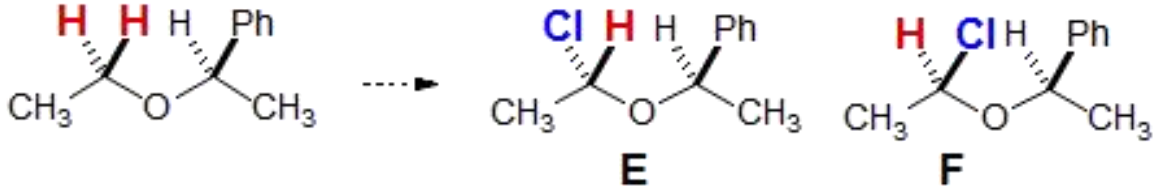
讨论一下：
这些化合物的核磁谱氢谱有几个峰？



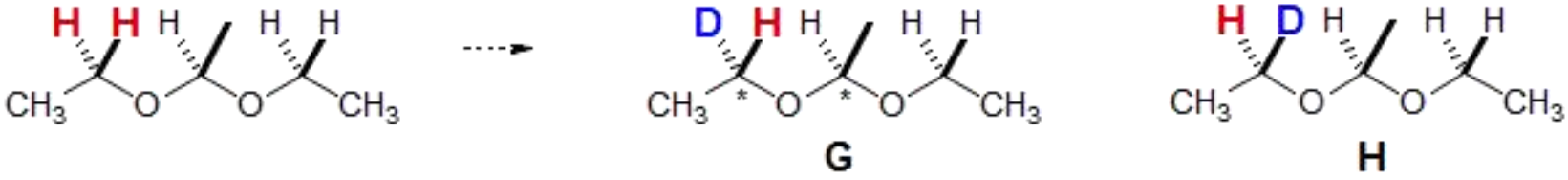
The structures **A** and **B** are **identical**, the two protons are **homotopic**.

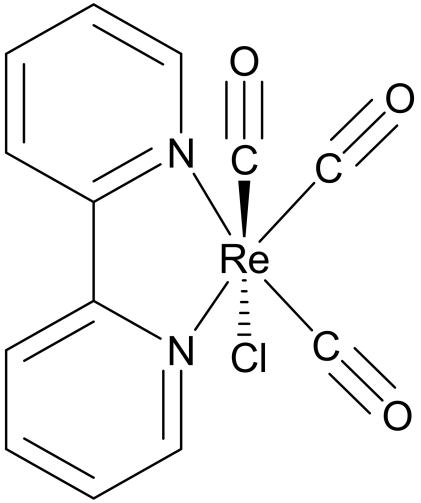


The structures **C** and **D** are **enantiomers**, the two protons are **enantiotopic**.



The structures **E** and **F** are **diastereomers**, the two protons are **diastereotopic**.





讨论一下：

1. 这个化合物是高自旋还是低自旋？
2. 它的核磁氢谱有几个峰？碳谱有几个峰？
3. 这个化学物的红外谱，有几个羰基伸缩振动峰？

群论初步

- 群论的基本理论和方法跟物质结构的**对称性**结合起来，是研究化学的一种有力工具.

- 群论在化学中的应用:

  - 从对称性的角度，描述分子的立体构型以及分子轨道结构
  - 定性推断组成杂化轨道的原子轨道
  - 电子能态在不同晶体场中的分裂情况，及电子跃迁的选律
  - 判断简正振动在红外(Infrared，IR)或Raman光谱中的活性、预言可能出现的谱带的数目

第一节、对称操作和对称元素

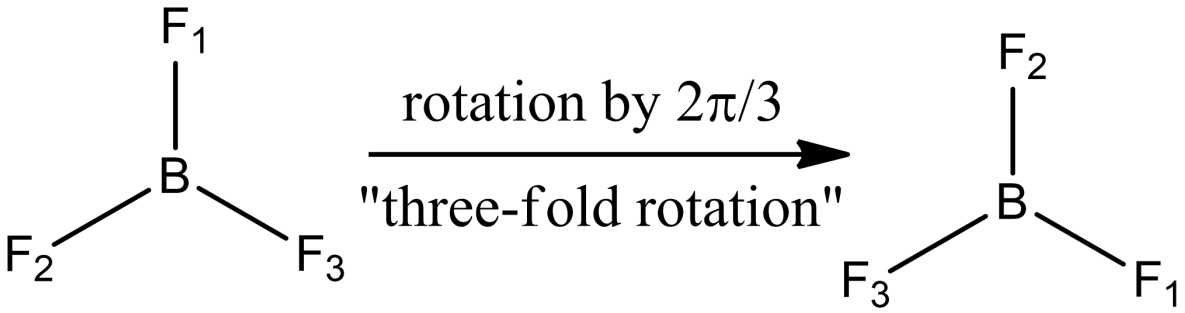
- **分子的对称性, 对称操作及对称元素**

  定义:**分子的对称性**是指存在一定的操作，它在保持任意两点间距离不变的条件下，使分子内部各部分变换位置，而且变换后的分子整体又恢复原状，这种操作称为**对称操作**(symmetry operation)．对称操作据以进行的元素称为**对称元素**(symmetry element).

举例：三氟化硼

Definitions:

Symmetry Operation: A symmetry operation is an operation which moves a molecule into a new position equivalent to its original one.





- 对称元素：
  - 旋转轴
  - 镜面



对称操作类型

- 旋转
- 反映
- 反演
- 旋转反映
- 恒等操作



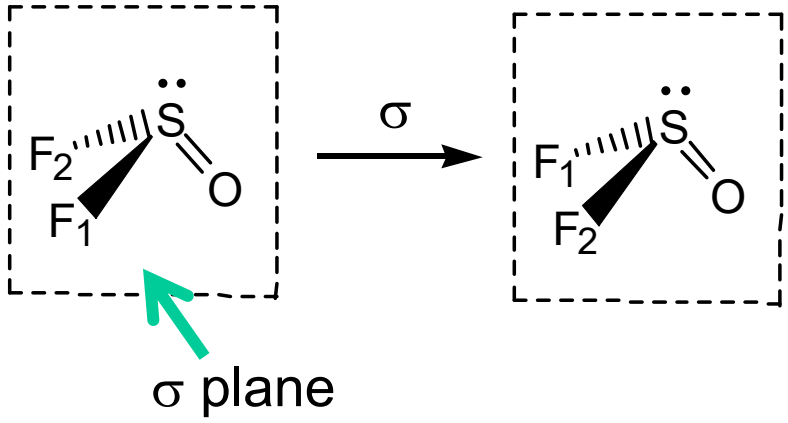








F_2SO: pyramidal structure



镜面

$\sigma \cdot \sigma = E$

Two successive mirror plane operations will have net zero effect on F_2SO.

The mirror plane passes through O and S, and bisects the FSF angle

Two operations whose product does not depend on the order of the multiplication (i.e., X•Y = Y•X) are said to commute

Note: X•Y means Y operates first.

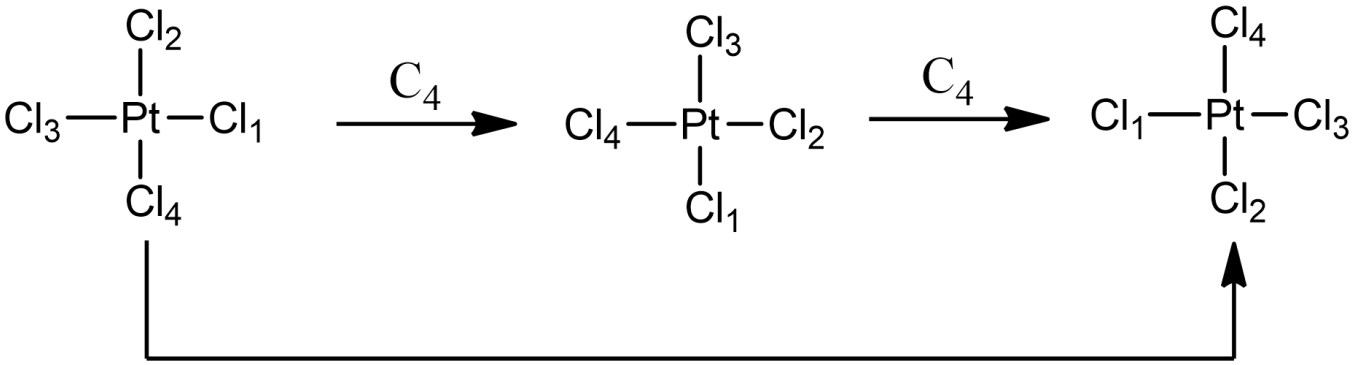
The product X•Y is NOT necessarily equal to Y•X

3) Rotation about an axis is denoted $\mathbf{C_n}$: C means rotation and n tells what fraction of a complete rotation through 2π we are to perform.

For BF_3: rotate 120 ° ($2\pi/3$) and it is called a C_3 rotation

For $PtCl_4^{2-}$:

旋转轴



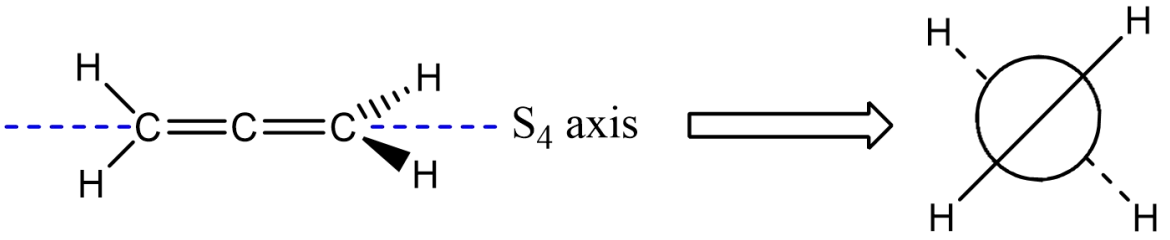
$$C_4 \cdot C_4 = C_4^2 = C_2$$

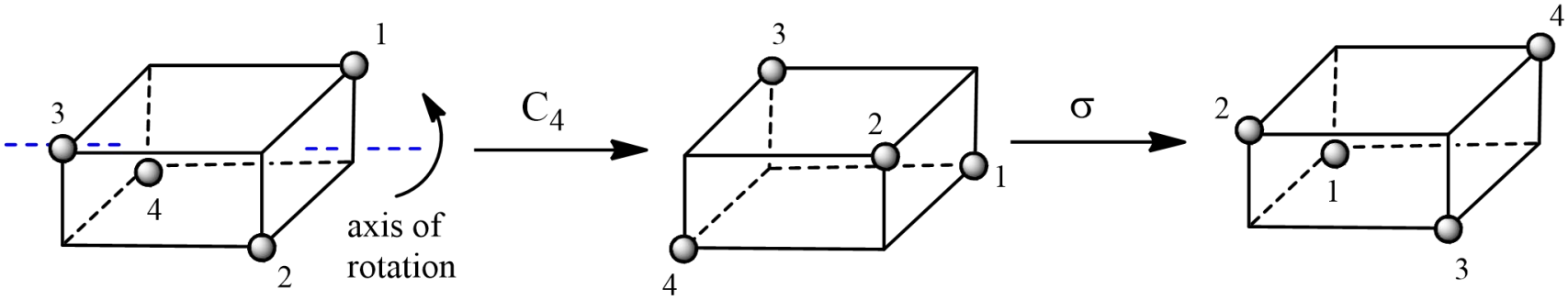
What is the inverse of C_4? $C_4^3 \bullet C_4 = E$

In general, the inverse of C_n^m is C_n^{n-m}

4) An improper rotation consists of a rotation followed by a reflection through the plane perpendicular to the axis of rotation

反轴

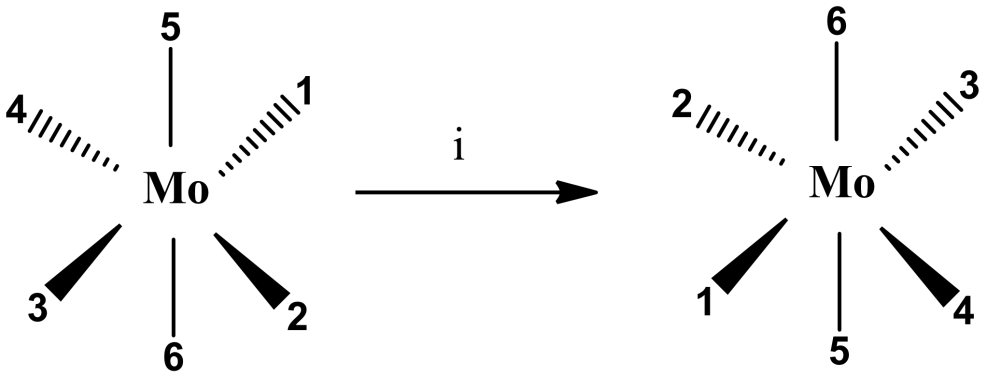




5) Inversion involves passing each atom through two center of the molecule and placing it on the opposite side of the molecule.

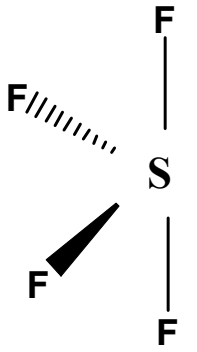
i=S2

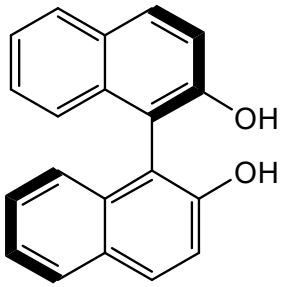
Mo(CO)6



对称中心

Examples: SF4





C_2 axis passing through mirror→the plane of the page

mirror →perpendicular to the page(colinear of the C_2 axis)

Group: The symmetry operator which apply to any particular molecule collectively possess the properties of a mathematical group.

群的定义

Every group has four properties:

1) These must exist an identity operator(E) which commutes with all other operators and leaves them unchanged.

2)The product of any two operators must also be a member of the group.

3)Multiplication is associative, which means that you may group the operation as you please, so long as you don't reverse the order of operations: $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

4)These must exist in the group an inverse (also called the reciprocal) for each other. The product of an operator and its inverse is the identity operator.

An operator and its inverse must commute.

$$Z \cdot Z^{-1} = Z^{-1} \cdot Z = E$$

A subgroup is a group within a group

第二节、对称操作群

目标：描述分子的对称性

- 定义：对称操作的集合构成的群称为对称操作群，简称对称群(symmetry group)

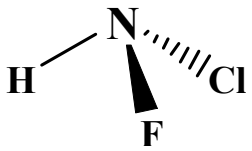
- 对称操作群也必具有数学上群的四条基本性质.

Point groups: It is possible to assemble symmetry operations into mathematical groups.

It is possible to assign any molecule to a point group.

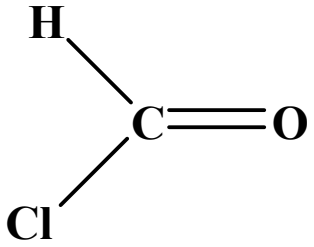
点群

Point Group C_1: A trivial group →molecules in this point group have no symmetry.



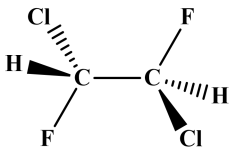
C1

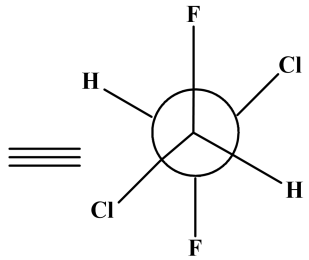
Point Group Cs: The only symmetry is a plane of reflection, formyl chloride:



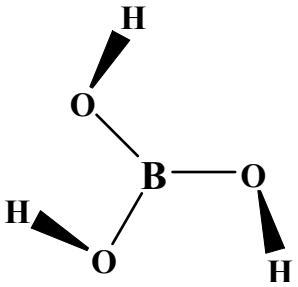
Molecular plane

Point Group Ci: molecules possessing only a center of inversion.





Point Group Cn: molecules possess only an n-fold axis of rotation boric acid:



Let's see if all the symmetry operations for Boric acid satisfy the requirement of a mathematical group.

a) E operation; **b)** the product of any two operations must also be a member of the group; **c)** associative property; **d)** inverse

a)And b) are satisfied.

C_3	E	C_3	C_3^2
E	E	C_3	C_3^2
C_3	C_3	C_3^2	E
C_3^2	C_3^2	E	C_3

c) $C_3 \bullet (C_3 \bullet C_3^2) = (C_3 \bullet C_3) \bullet C_3^2$

$\| \qquad \|$

$C_3 \bullet E \qquad C_3^2 \bullet C_3^2$

$\| \qquad \|$

$C_3 \qquad C_3$

⇩

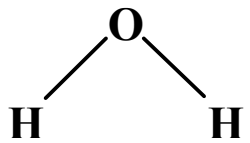
yes

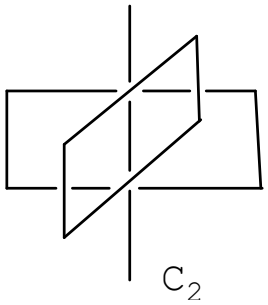
d) Inverse of E is E
inverse of C_3 is C_3^2
inverse of C_3^2 is C_3

The symmetry operations for a C_3 point group do conform to the requirements of a mathematical group.

Point Group C_{nv}: A molecule in C_{nv} point group possesses a C_n asis and n vertical planes that are colinear with the C_n axis. .

H_2O





C_{2v}:

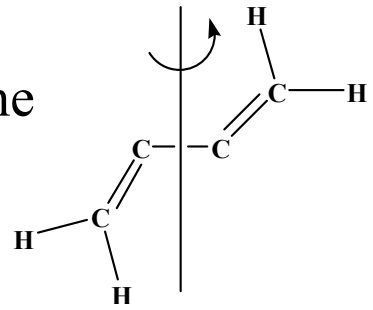
C_{2v}	E	C_2	σ_v	σ_v'
E	E	C_2	σ_v	σ_v'
C_2	C_2			
σ_v	σ_v			
σ_v'	σ_v'			

→

C_{2v}	E	C_2	σ_v	σ_v'
E	E	C_2	σ_v	σ_v'
C_2	C_2	E	σ_v'	σ_v
σ_v	σ_v	σ_v'	E	C_2
σ_v'	σ_v'	σ_v	C_2	E

Point Group C_{nh}: generated by a C_n axis and a horizontal mirror plane σ_h .(σ_h is perpendicular to C_n)

Trans-butadiene



Plus mirror in the molecular plane

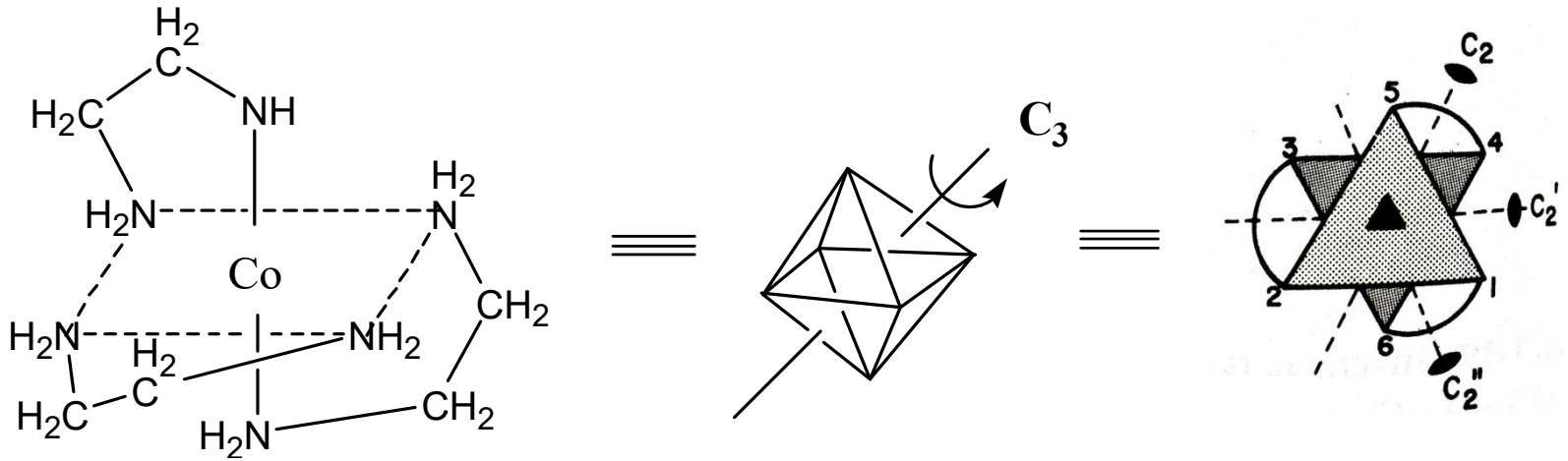
$C_2 \bullet \sigma_h = S_2 = i$ ⟶ An inversion center introduced by the product of C_2 and σ_h

Multiplication Table for C_{2h}

C_{2h}	E	C_2	σ_h	i
E	E	C_2	σ_h	i
C_2	C_2	E	i	σ_h
σ_h	σ_h	i	E	C_2
i	i	σ_h	C_2	E

Point Group D_n: A D_n point group is generated by a C_n axis and a C_2 axis perpendicular to the C_n axis

$Co(H_2NCH_2CH_2NH_2)_3^{3+}$



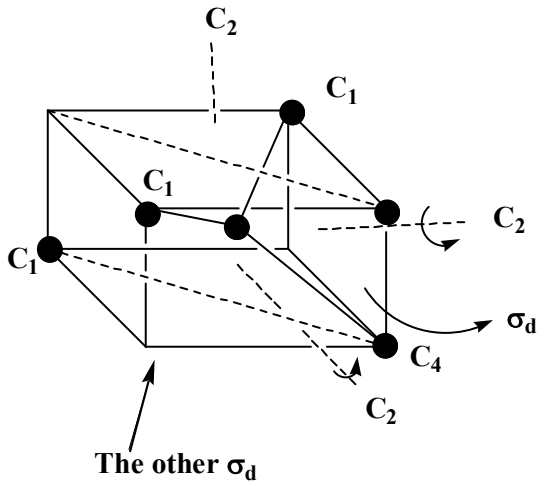
Note: A C_n axis and one perpendicular C_2 axis will generate (n-1)C_2 axis.

C_3 axis is called the principal axis C the axis of the highest order in a point group.

Point Group D_{nd}: generated by a C_n axis, a perpendicular C_2 axis, and a dihedral mirror plane,σ_d.

The σ_d plane is colinear with respect to The principal axis, but bisects $\perp C_2$ planes.

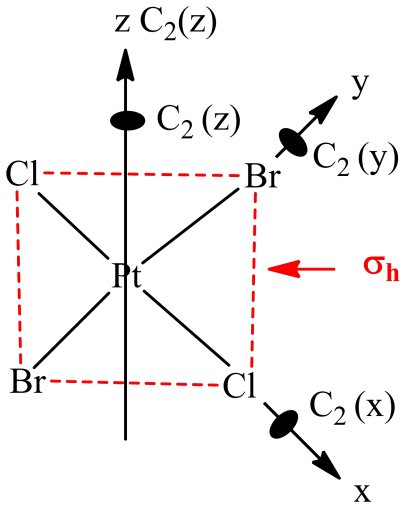
Cs_2CuCl_4---a squashed tetrahedral Pzd structure



Point Group D_{nh}: a C_n axis, n $\perp C_2$ axis, and a horizontal mirror plane that is perpendicular to the principal axis.

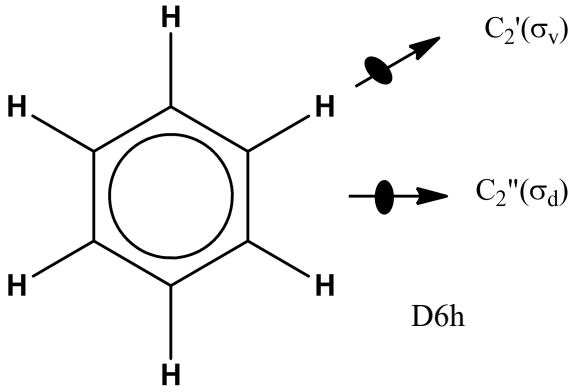
*These symmetry elements will generate n vertical mirror planes as well.

trans-$PtCl_2Br_2^{2-}$



*plus two vertical mirror planes along Cl-Pt-Cl and Br-Pt-Br lines, respectively

Benzene

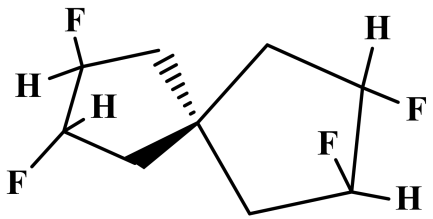


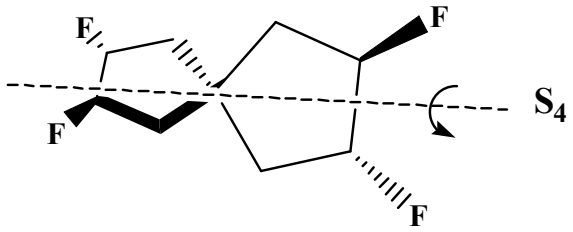
C_2',σ_v colinear and pass through as many atoms as possible;
σ_2", σ_d colinear and pass through as few atoms as possible.

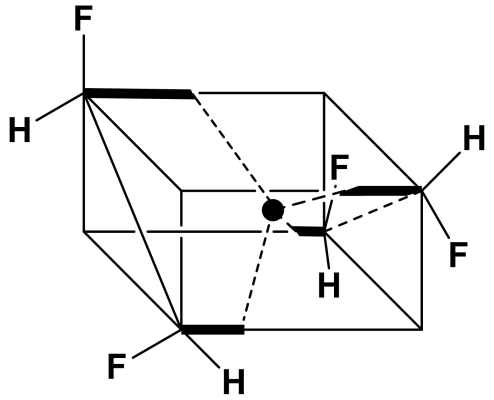
Two other set σ_2' & σ_v/ C_2"& σ_d

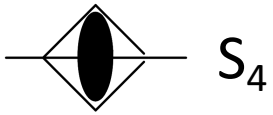
Point Group S_n: generated by a S_n

Tetrafluorospirononane:









S_4,

$S_4^2=C_2$,

S_4^3

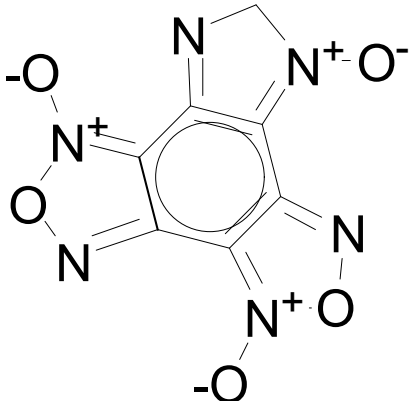
$S_4^4=E$

$S_2=i$

When n=odd#, $S_n=C_{nh}$

$\therefore$ only S_4, S_6, S_8……are unique point groups.

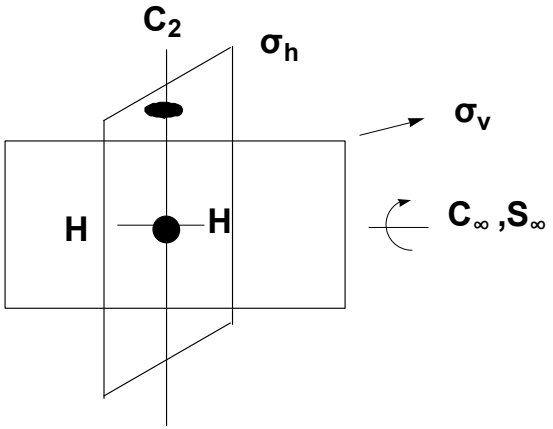
Benzo trifuroxan:



$C_{3h}=S_3$

Special Point Groups:

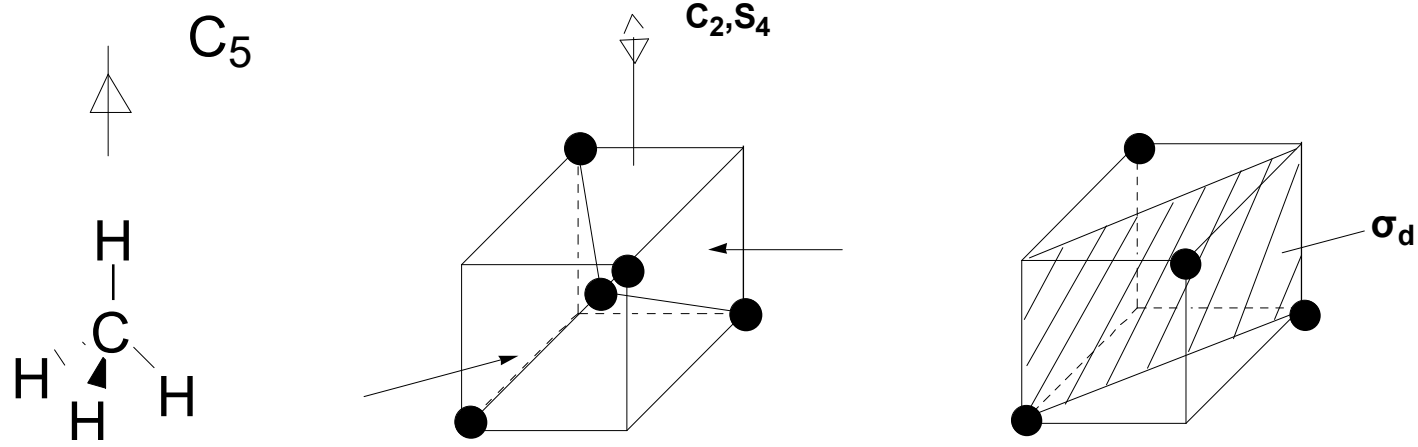
$D_{\infty h}$: linear molecule with a center of inversion.



H_2 → a C axis of infinite order
an infinite number of perpendicular C_2 axis.
an infinite number of vertical mirror planes
a horizontal mirror plane(σ_h)
an inversion center at the center of the molecule.
(C_∞ axis is also a S_∞ axis)

$C_{\infty v}$: linear molecules without an inversion center such as heteronuclear diatomic molecules.(HCl).

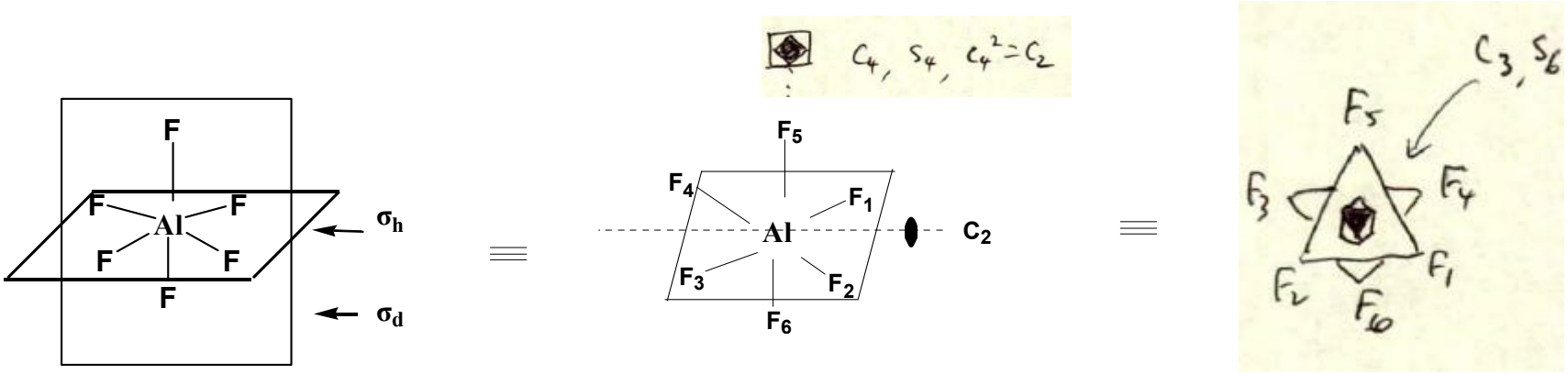
T_d: tetrahedral



24 operations in T_d point group with symmetry elements E,C_3, C_2, S_4 and σ_d

Note: no inversion center for T_d point group.

O_h: octahedral molecules such as AlF_6^{3-}



O_h group contains C_4,C_5,C_2,S_4 and S_4 axis, as well as σ_h and σ_d mirror planes, and an inversion center.

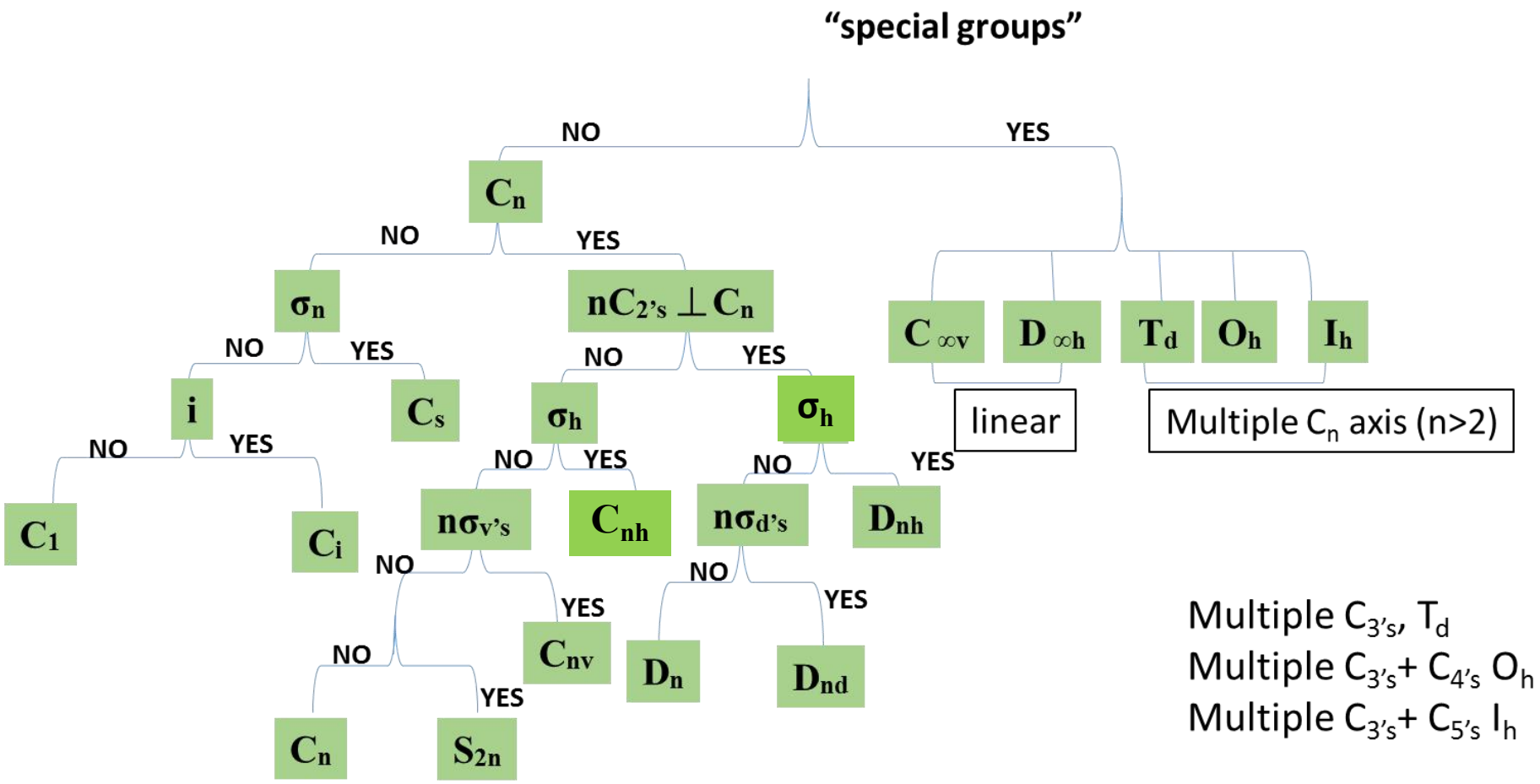
All of octahedral and tetrahedral point groups are called cubic groups.

Symmetry of Point Groups

point group	Important sym elements	order of the gourp
C_1	E	1
C_i	i	2
C_s	σ	2
C_n	C_n	n
S_{2n}	S_{2n}	2n
C_{nv}	C_n, σ_v	2n
C_{nh}	C_n, σ_h	2n
D_n	C_n, $\perp C_2$	2n
D_{nd}	C_n, $\perp C_2$, σ_d	4n
D_{nh}	C_n, $\perp C_2$, σ_h	4n
$C_{\infty V}$	linear molecules w/o i	∞
$D_{\infty h}$	linear molecules w/. i	∞
T_d	tetrahedral symmetry	24
T_h	tetrahedral symmetry,σ_h	24
O_n	octahedral symmetry	48
I_n	icosahedral symmetry	120
K_n	spherical symmetry	∞

点群的判定步骤

Point Group Classification:



Multiple $C_{3's}$, T_d
Multiple $C_{3's}$+ $C_{4's}$ O_h
Multiple $C_{3's}$+ $C_{5's}$ I_h

第三节、对称操作的表示矩阵

目标：描述一个对象在对称操作下变换的对称性

看到这一页感到不适的同学请关闭该ppt直接打开下一部分的ppt

- 笛卡尔坐标系中，物体上的任一点的坐标为x、y、z，对称操作使该点的坐标发生变换．因此，对称操作的作用结果相当于不同的坐标变换.
- 坐标变换可以用矩阵表示．换句话说，对称操作可以用矩阵来表示.
- 若存在一组坐标的函数，当坐标变换时，其中的任一函数变为这组函数的一个线性组合，故由对称操作导致的这组函数的变化情况也可以用矩阵来表示.
- 描述各种对称操作作用结果的矩阵称为表示矩阵.
- 表示矩阵既可以从对称操作作用下任意点的坐标的变换情况得到，也可以从一组适当的函数得到，这组函数称为相应表示矩阵的基函数.
- 选择不同的基函数，对称操作的表示矩阵不同.

Matrix Representation of Symmetry Operations:

数学复习：向量和矩阵是什么？

Vector: a series of numbers in a row or a column.

Example: (1 3 12 e cos2π/17) a row vector

$$\begin{pmatrix} 1 \\ 3 \\ 12 \\ e \\ \cos 2\pi/17) \end{pmatrix}$$

A column vector

Matrix: any rectangular array of numbers set between two brackets

square matrix: with equal numbers of rows and columns

Element a_{ij}: ith row and jth column

$$\mathbf{A}= \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ . & . & . & \dots & \\ . & . & . & \dots & \\ . & . & . & \dots & \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix}$$

The product of a square matrix and a vector:

数学复习：矩阵和向量的积

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ . & . & . & \dots & \\ . & . & . & \dots & \\ . & . & . & \dots & \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix} \bullet \begin{pmatrix} x_1 \\ x_2 \\ . \\ . \\ . \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ . \\ . \\ . \\ y_n \end{pmatrix}$$

$\mathbf{y_m=\sum_j a_{mj}x_j}$ e.g. $y_1=\Sigma a_{1j}x_j=a_{11}x_1+a_{12}x_2+\dots+a_{1n}x_{1n}$

Example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \bullet \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1\bullet4+2\bullet5+3\bullet6 \\ 0\bullet4+2\bullet5+1\bullet6 \\ 1\bullet4+0\bullet5+2\bullet6 \end{pmatrix} = \begin{pmatrix} 32 \\ 16 \\ 16 \end{pmatrix}$$

The product of two square matrices:

数学复习：矩阵的积

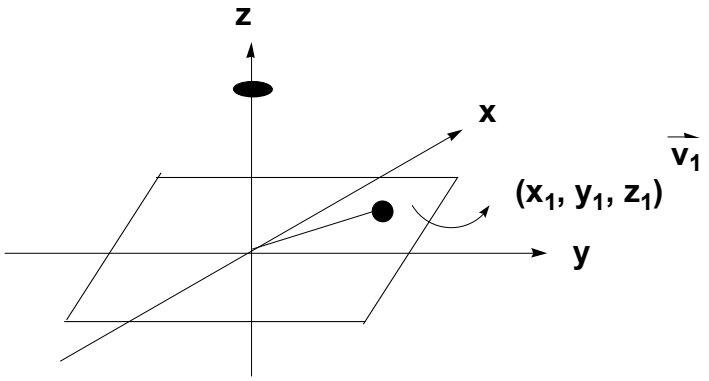
$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \cdot & \cdot & \cdot & \dots & \\ \cdot & \cdot & \cdot & \dots & \\ \cdot & \cdot & \cdot & \dots & \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix} \cdot \begin{pmatrix} \overrightarrow{x_1} & \overrightarrow{x_2} & & \overrightarrow{x_n} \\ x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & \cdots\cdots & & \\ & \cdot & & \\ & \cdot & & \\ x_{n1} & \cdots\cdots & & x_{nn} \end{pmatrix} = \begin{pmatrix} \overrightarrow{y_1} & \overrightarrow{y_2} & & \overrightarrow{y_n} \\ y_{11} & y_{12} & \dots & y_{1n} \\ y_{21} & \cdots\cdots & & \\ & \cdot & & \\ & \cdot & & \\ y_{n1} & \cdots\cdots & & y_{nn} \end{pmatrix}$$

$$A \cdot X = Y$$

Example:

$$\begin{pmatrix} 0 & 1 & 2 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 4 & 5 \\ 5 & 9 & 13 \\ 1 & 4 & 7 \end{pmatrix}$$

C_{2h}



对称操作的矩阵表示：用具体的数学表示抽象的操作

基组

表示矩阵

$$E \bullet \vec{\nu}_1 = \vec{\nu}_1 \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$i \bullet \vec{\nu}_1 = -\vec{\nu}_1 \qquad \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} -x_1 \\ -y_1 \\ -z_1 \end{pmatrix}$$

$$\sigma \bullet \vec{\nu}_1 = \vec{\nu}_2 \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ -z_1 \end{pmatrix}$$

R_θ : a rotation through the angle θ

$$\mathbf{R_\theta} \cdot \vec{\mathbf{r}_1} = \vec{\mathbf{r}_2}$$

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

C_2 operation does not change z coordinate

这些表示矩阵满足与对称操作相似的乘法，组成一个群

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} -x_1 \\ -y_1 \\ z_1 \end{pmatrix}$$

$$\mathbf{C_2} \cdot \vec{\nu}_1 = \vec{\nu}_3$$

It is super easy to probe:

$$\mathbf{C_2} \cdot \mathbf{\sigma_h} = \mathbf{i}$$

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Each matrix has an inverse matrix. The identity matrix I

$$Q \bullet Q^{-1} = Q^{-1} \bullet Q = I$$

(a matrix and its inverse commute)

Similarity Transformation:

相似变换=改变基组

$$B = Q^{-1} \bullet A \bullet Q$$

A and B are conjugate

$$B = Q^{-1} \bullet A \bullet Q$$

$$A \bullet X = Y$$

$$(Q \bullet B \bullet Q^{-1}) \bullet X = Y$$

$$B \bullet (Q^{-1} \bullet X) = (Q^{-1} \bullet Y)$$

Characters of conjugate matrices

相似变换=改变基组

Character: the sum of the diagonal elements in a square matrix

$$\chi_A = \Sigma_j a_{jj}$$

a) The character of A•B= the character of B•A

b) Conjugate matrices have identical characters

…

Reducible vs. Irreducible Representations:

A, B, C…represent the various operations of a point group:

$$A' = Q^{-1} \bullet A \bullet Q$$
$$B' = Q^{-1} \bullet B \bullet Q$$
$$C' = Q^{-1} \bullet C \bullet Q$$

This similarity transformation does not change the character of a matrix, and is equivalent to a change of basis

相似变换=改变基组

目标：描述一个对象在对称操作下变换的对称性

C_{2v}	E	C_2	σ_v	$\sigma_{v'}$
Γ_1	1	1	1	1
Γ_2	1	-1	-1	1
Γ_3	1	-1	1	-1
Γ_4	1	1	-1	-1
…	…	…	…	…

不同基组

迹

相似变换=改变基组，但不是所有的基组之间都可以通过相似变换建立联系。有一些基组下矩阵的迹不同。这些基组之间有本质差别。他们所对应的矩阵表示也是有本质差别的表示

可以无限地选择基组，得到无限种矩阵的表示。但是大部分表示之间是本质相同的，可由相似变换联系。本质不同的表示一共有有限多种，他们就是不可约表示

第四节、特征标表的构建

目标：构建特征标表

Great Orthogonality Theorem

广义正交定理

$$\sum_R [\Gamma_i(R)_{mn}][\Gamma_j(R)_{m'n'}]^* = \frac{h}{\sqrt{l_i l_j}} \delta_{ij}\delta_{mm'}\delta_{nn'}$$

$\Gamma_i(R)$ --- Matrix that represents the operation R in the ith representation. Its form can depend on the basis for the representation.

$\Gamma_i(R)_{mn}$ --- Matrix element in the mth row and n^{th} column of $\Gamma_i(R)$

l_i --- The dimension of the ith representation

$$\sum_i l_i^2 = h$$

h --- The order of the group(the number of operations)

δ_{ij} = 1 If i=j, 0 otherwise

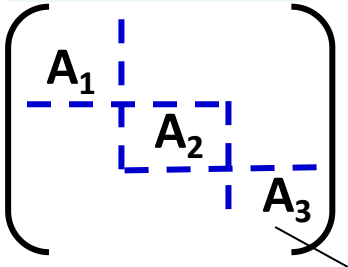
以C_{3v}群为例

不同基组

i

C_{3v}	E	$2C_3$	$3\sigma_v$	基
A_1	1	1	1	z, z^2
A_2	1	1	−1	
E	2	−1	0	(x, y), (xz, yz)

具体矩阵



$\Gamma_i(R)$

$\Gamma_i(R)_{mn}$

表示矩阵的迹

$$\sum_i l_i^2 = h$$

$$\sum_R [\Gamma_i(R)_{mn}][\Gamma_j(R)_{m'n'}]^* = \frac{h}{\sqrt{l_i l_j}} \delta_{ij} \delta_{mm'} \delta_{nn'}$$

1. Characters of different irreducible representations are orthogonal:

The vectors whose components are the characters of two different irreducible representations are orthogonal, that is

$$\sum_R \chi_i(R)\cdot \chi_j(R) = 0 \quad when\ i \neq j$$

$$= \text{h} \quad \text{when i=j}$$

广义正交定理的三个推论

2. # of classes = # of irreducible representatives

3. $\sum_i l_i^2 = h$

h is The order of the group (the number of operations)

l_i is The dimension of the ith representation

例：利用这些数学定理，求不可约表示 C_{2v}

- Four symmetry elements, and each of them in a separate class→four irreversible representations
- $l_1^2 + l_2^2 + l_3^2 + l_4^2 = 4$
- The only solution is $l_1 = l_2 = l_3 = l_4 = 1$
- So C_{2v} has four 1-D irreversible representations
- The first one is easy

C_{2v}	E	C_2	σ_v	$\sigma_{v'}$
Γ_1	1	1	1	1

- They must be orthogonal to each other, so each one should have two 1's and two -1's

C_{2v}	$\boldsymbol{E}$	$\boldsymbol{C}_2$	σ_v	$\sigma_{v'}$
Γ_1	1	1	1	1
Γ_2	1	-1	-1	1
Γ_3	1	-1	1	-1
Γ_4	1	1	-1	-1

例：利用这些数学定理，求不可约表示 C_{3v}

- Six symmetry elements, but three classes →three irreversible representations
- $l_1^2 + l_2^2 + l_3^2 = 6$
- The only solution is $l_1 = l_2 = 1 \;\; l_3 = 2$
- So C_{2v} has two 1-D irreversible representations and one 2-D representations
- The first one is easy $1^2+2*1^2+3*1^2=6$

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ_1	1	1	1

- The second one is orthogonal to the first one , so each one should have two 1's and two -1's

$1^2+2*1^2+3*(-1)^2 = 6$

$1*1+2*1*1+3*1*(-1)=0$

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ_1	1	1	1
Γ_2	1	1	-1

Γ_3

$$\sum_R \chi_1(R)\cdot \chi_3(R) = 1\cdot 2 + 2\cdot 1\cdot \chi_3(C_3) + 3\cdot 1\cdot \chi_3(\sigma_v) = 0$$

$$\sum_R \chi_2(R)\cdot \chi_3(R) = 1\cdot 2 + 2\cdot 1\cdot \chi_3(C_3) + 3\cdot (-1)\cdot \chi_3(\sigma_v) = 0$$

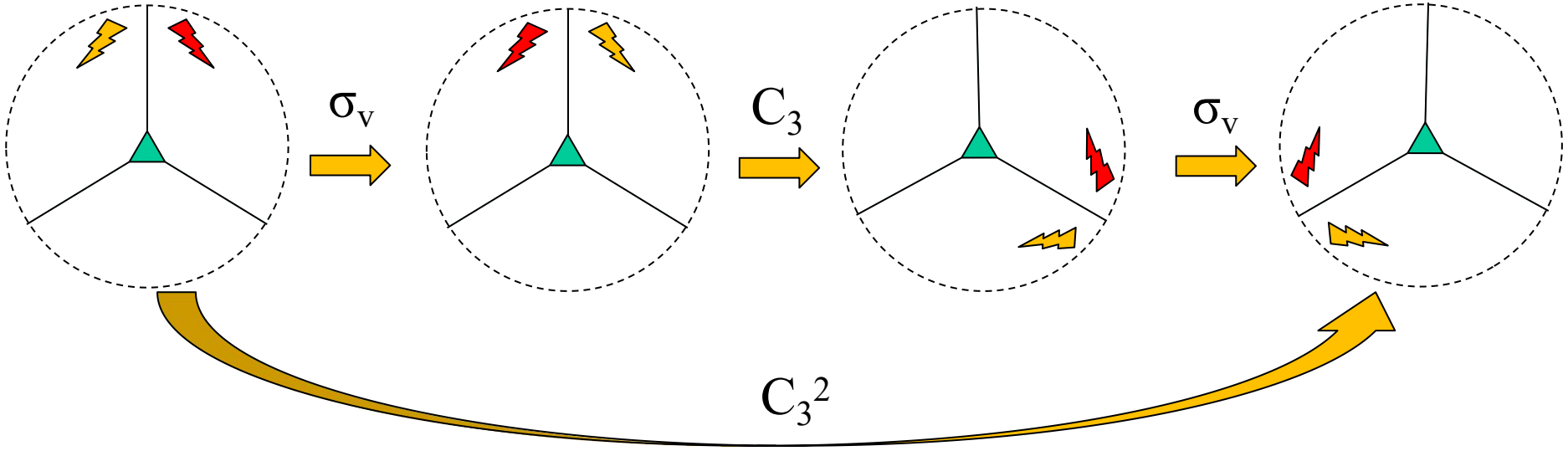
$$\chi_3(C_3) = -1 \qquad \chi_3(\sigma_v) = 0$$

C_{3v}	E	$2C_3$	$3\sigma_v$
Γ_1	1	1	1
Γ_2	1	1	-1
Γ_3	2	1	0

Why does C3v point group have three classes?

Reciprocal of each operation:

$E^{-1} = E$, $C_3^{-1} = C_3^2$, $(C_3^2)^{-1} = C_3$, $\sigma_v^{-1}=\sigma_v$, $\sigma_v'^{-1}=\sigma_v'$, $\sigma_v''^{-1}=\sigma_v''$



$\sigma_v^{-1} \cdot C_3 \cdot \sigma_v = C_3^2$

凡是能被其他对称操作联系的对称操作，属于同一class，有同样的特征标